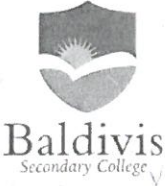


Name:	<u>Solutions</u>	
Class:		
	Year 8 Mathematics 2019	20/20 +5 for completed neat part 1
	Investigation #1 – Part 2 Validation	
	Perimeter and Area	
	Resources Allowed: Personal notebook/calculator/ Part 1 / Field Design	
	Time allowed: 1 full class lesson	

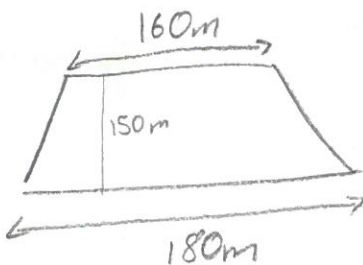
For Part 2 validation you will be using the given field design. Part 1 will help guide you through the process but only use the measurements in the given field design.

Question 1: Area

7 marks

Find the area of each of the following shapes from the given field design.

Total field area within the boundary lines:

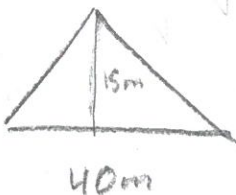


finding a and b

$$\frac{1}{2} \times (160 + 180) \times 150 \checkmark$$

$$= 25,500 \text{ m}^2 \checkmark$$

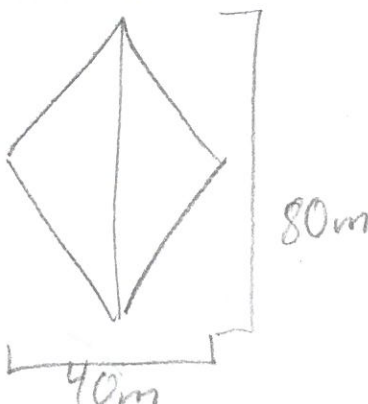
Inner centre:



$$\frac{1}{2} \times 15 \text{ m} \times 40 \text{ m} \checkmark$$

$$= 300 \text{ m}^2 \checkmark$$

Outer centre area:



$$\frac{1}{2} \times 40 \text{ m} \times 80 \text{ m} \checkmark$$

$$= 1,600 \text{ m}^2 \checkmark$$

Question 2: Line markings**8 marks**

The umpires have agreed upon this design and now need to change all 17 AFL grounds. They have been quoted by a turf painting company that it will cost \$0.32 to paint each metre of the new lines on the footy grounds. What will the total cost be for all 17 footy grounds to have the new lines painted?

$$\text{Boundary} = 170 + 160 + 170 + 180 = 680\text{m} \checkmark$$

$$2 \times 50\text{m lines} = 170\text{m} + 170\text{m} = 340\text{m} \checkmark$$

$$\text{Outer center} = 45 + 45 + 65 + 65 = 220\text{m} \checkmark$$

$$\text{Inner center} = 25 + 15 + 40 = 80\text{m} \checkmark$$

$$\text{All lines together} = 680 + 340 + 220 + 80 = 1,320\text{m}$$

$$1,320\text{m} \times 17 \times 0.32 = \$7,180.80$$

✓
multiplying
by 17

✓
multiplying
by \$0.32

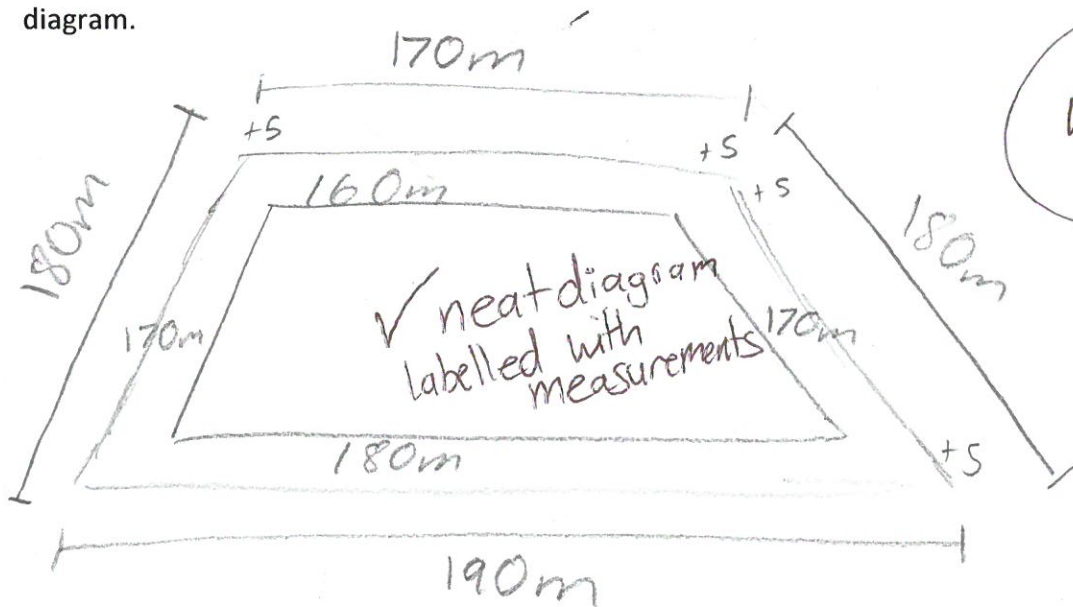
✓
Final answer

✓
(adding all
lines together)

-0.5 if no
\$

Question 3: Safety fencing**5 marks**

For safety reasons there needs to be 5m between the boundary lines of the field and the fence which spectators must stay behind so as not to get injured. How much fencing is required for this field? Draw a diagram.



✓
Finding
new
measurements
correctly

$$170\text{m} + 180\text{m} + 190\text{m} + 180\text{m} = 720\text{m of fencing}$$

✓
adding up the sides